

Section 33: Cisco Device Security

Layer: Multi | Reference: Odom Vol2 Ch4-5: Securing Network Devices, Local AAA | *Original study notes — CCNA 200-301*

Hardening the management plane of routers and switches - how administrators access and authenticate to devices - is foundational to overall network security and a direct CCNA exam topic.

Key Concepts

- Telnet sends all traffic, including credentials, in plaintext - SSH encrypts the session and is the expected standard for remote management
- SSH requires a hostname, a domain name, and a generated RSA key pair (minimum 768 bits, commonly 2048+ recommended) before it can be enabled
- 'transport input ssh' on VTY lines disables Telnet entirely, forcing SSH-only remote access
- Local AAA (Authentication, Authorization, Accounting) uses a local username/password database stored on the device itself - simple but doesn't scale across many devices
- Centralized AAA via RADIUS or TACACS+ servers allows credential management from one place across the whole network instead of per-device local accounts
- TACACS+ (Cisco-favored) separates authentication, authorization, and accounting into distinct steps and encrypts the entire packet; RADIUS (open standard) combines authentication and authorization and only encrypts the password field
- Banner motd (message of the day) displays a legal warning banner before login - courts have used the presence/absence of such banners in unauthorized-access prosecutions, so this is a genuine best practice, not just decoration
- Privilege levels (0-15) on IOS allow granular access control; level 1 = user EXEC, level 15 = full privileged EXEC, with custom levels possible in between for delegated admin tasks
- Role-Based Access Control (RBAC) / CLI views let you create custom named views exposing only specific commands to specific admin roles
- Console, AUX, and VTY lines should each have authentication configured - an unsecured console or AUX port is a common physical-access attack vector

Command Reference

Command	Purpose
<code>ip domain-name <name></code>	Required before generating an RSA key for SSH
<code>crypto key generate rsa modulus <bits></code>	Generate the RSA key pair required to enable SSH
<code>ip ssh version 2</code>	Force the device to use SSH version 2 only (more secure than v1)
<code>line vty 0 4 / transport input ssh</code>	Restrict VTY remote access to SSH only, disabling Telnet

Command	Purpose
username <name> privilege <level> secret <password>	Create a local user account with a specific privilege level
aaa new-model	Enable the AAA subsystem before configuring authentication/authorization methods
tacacs server <name> / radius server <name>	Define a centralized AAA server
banner motd #<message>#	Configure a login warning banner (delimiter character framing the message)
show ip ssh	Verify SSH version and status

Configuration Walkthrough

Enable SSH-only access and disable Telnet

```
R1(config)#ip domain-name example.com
R1(config)#crypto key generate rsa modulus 2048
R1(config)#ip ssh version 2
R1(config)#username admin privilege 15 secret StrongPass123
R1(config)#line vty 0 4
R1(config-line)#transport input ssh
R1(config-line)#login local
```

Configure a legal warning banner

```
R1(config)#banner motd #
Unauthorized access to this device is prohibited.
#
```

Enable centralized AAA with a TACACS+ server as primary, local as fallback

```
R1(config)#aaa new-model
R1(config)#tacacs server TACACS1
R1(config-server-tacacs)#address ipv4 10.10.10.20
R1(config-server-tacacs)#key SharedSecret123
R1(config)#aaa authentication login default group tacacs+ local
```

Worked Scenario

Q: An administrator can reach a router via Telnet but SSH connection attempts are refused. They've already set a hostname and domain-name. What's missing?

Almost certainly the RSA key pair was never generated, or VTY lines still have 'transport input telnet' (or 'all') instead of 'ssh'. SSH on IOS requires hostname + domain-name + a generated crypto key (the key generation step itself enables the SSH server process); without 'crypto key generate rsa', SSH never actually starts even if everything else looks right.

Common Mistakes / Gotchas

- Forgetting that SSH requires a domain-name AND a generated RSA key, not just enabling the service
- Leaving 'transport input telnet' or 'transport input all' configured, which keeps insecure Telnet available even after enabling SSH
- Configuring AAA with no fallback method, risking total lockout if the TACACS+/RADIUS server becomes unreachable
- Using only 'enable password' or 'enable secret' without per-user accountability via local usernames or centralized AAA

Exam Tips

- ★ *Know the exact SSH enablement sequence: hostname -> domain-name -> crypto key generate rsa -> (optionally) ip ssh version 2*
- ★ *TACACS+ encrypts the whole packet and separates AAA into 3 distinct phases; RADIUS only encrypts the password and combines authentication+authorization*
- ★ *Always configure a fallback authentication method when using centralized AAA, to avoid total lockout*